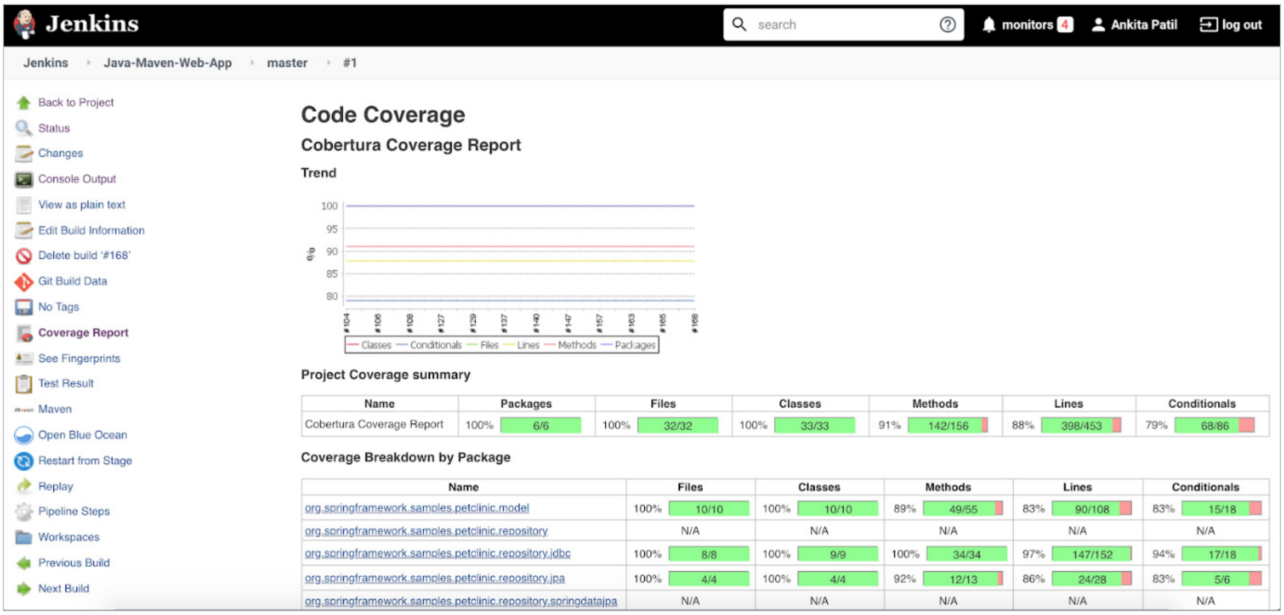

Figure 4: Cobertura code coverage on Jenkins

```
pipeline {
  agent {
    node {
      label 'master'
    }
  }
  stages {
    stage('Continuous Integration') {
      steps {

        withMaven(jdk: 'JAVA_HOME', maven: 'MAVEN_HOME') {
          bat 'mvn clean'
          bat(script: 'mvn test cobertura:cobertura install',
label: 'Unit Testing and Code Coverage')
          cobertura(autoUpdateHealth: true,
autoUpdateStability: true, classCoverageTargets: 'target/
site/cobertura/', coberturaReportFile: 'target/site/
cobertura/*.xml', failUnstable: true, zoomCoverageChart:
true)
        }

        withSonarQubeEnv(installationName: 'SonarQube-
Server', credentialsId: 'SonarToken') {
          bat(script: 'D://Softwares//sonar-scanner-cli//
sonar-scanner-4.3.0.2102-windows//bin//sonar-scanner
-Dproject.settings=sonar-project.properties', label:
'SonarQube Analysis')
        }

        waitForQualityGate(abortPipeline: true,
credentialsId: 'SonarToken', webhookSecretId: 'SonarWebHook')
      }
    }
  }
}
```

The code has referenced the SonarQube server (SonarQube-Server), SonarToken and SonarWebhook, which we configured in the 'Manage Jenkins with Jenkins and SonarQube' setup.

The two steps performing SonarQube analysis in *Jenkinsfile* above are:

1. *withSonarQubeEnv*: SonarQube analysis using sonar-scanner in batch script and the *sonar-project.properties* files we created.
2. *waitForQualityGate*: Checking for the 'quality gate' status after the SonarQube analysis is executed.

Save the pipeline and run it again. On its successful execution, we will be able to find 'quality gate' results in the Jenkins pipeline console, as shown in Figure 5.

The SonarQube analysis result on the SonarQube server looks like what's shown in Figure 6.

We have now successfully analysed the Maven Java application using SonarQube analysis in Jenkins CI pipeline.

Understanding the SonarQube analysis report from its project dashboard

Once we analyse the source code and the result is published on the SonarQube server, we can see the entire report of issues, code coverage and duplications on a single dashboard.

Let us first understand the different kinds of issues we see in Figure 6.